Mathematical Foundations for Power Analysis in Complex Systems

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Abstract

Mathematical Framework for Power Analysis. Understanding societal risks—from institutional capture to AI-enabled power concentration —requires rigorous models of how power operates between agents with complex, evolving relationships. I develop a formal mathematical framework for power dynamics by extending Dahl's influential 1957 theory, addressing its fundamental limitations through game-theoretic foundations.

Key Contributions. First, I demonstrate why absolute power models are insufficient, proving that relational power frameworks form a proper superset of absolute ones. Second, I show that Dahlian power cannot handle influence chains or compare power across different domains—critical limitations for analyzing complex systems. Third, I establish that utility functions are necessary for coherent power analysis, formalizing when power becomes behaviorally meaningful. This framework provides theoretical foundations for analyzing dangerous power dynamics in contexts where agents may lack clear utility functions or act irrationally over extended periods.

1. Introduction

Understanding power dynamics is critical for analyzing societal risks, from institutional capture to AI-enabled concentration of control. Yet despite decades of research across sociology, political science, and economics, we lack rigorous mathematical frameworks for modeling how power operates in complex multi-agent systems.

Consider these scenarios: A social media algorithm subtly shapes millions of political opinions through content curation. A powerful corporation influences regulatory policy through a web of lobbying relationships. An AI system gradually assumes decision-making authority across critical infrastructure. Each involves intricate power relationships that existing theories struggle to model precisely.

The Challenge. Current approaches to power analysis suffer from fundamental limitations. Sociological theories offer rich conceptual frameworks but lack mathematical precision needed for quantitative analysis. Game-theoretic models provide formal rigor but assume rational agents with well-defined utility functions—assumptions that often fail in real-world power dynamics where agents act irrationally, preferences evolve, or utility functions remain unclear.

Our Approach. This paper develops a formal mathematical framework for power analysis by extending Dahl's seminal 1957 theory of power. Dahl's framework – "A has power over B to the extent that A can get B to do something B would not otherwise do"—provides an intuitive starting point but faces severe limitations when analyzing complex systems.

Contributions. We make three key theoretical contributions. First, we prove that models of relative power form a proper superset of absolute power models, establishing why relational frameworks are necessary. Second, we demonstrate that Dahlian Power cannot handle influence chains or enable cross-domain power comparisons – critical limitations for analyzing modern power structures. Third, we establish formal conditions under which utility functions become necessary for coherent power analysis.

Roadmap. §2 introduces our formal modeling approach and contextualizes it within existing power literature. §3 formalizes and extends Dahl's framework, revealing its structural limitations. §4 proves the inadequacy of absolute power models. §5 demonstrates why utility functions are essential for meaningful power analysis. Together, these results provide theoretical foundations for analyzing dangerous power dynamics in complex systems.

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2. Towards a Formal Model of Power

The Literature Landscape. Power theory suffers from conceptual fragmentation. Existing approaches fall into two problematic categories: either loosely connected sociological definitions that resist formalization, or rigorous mathematical models confined to narrow domains (e.g., cooperative game theory's power indices).[1] Each theoretical tradition critiques its predecessors while offering new definitions, creating a tower of competing frameworks rather than cumulative progress.[2]

Contemporary Applications. Formal power modeling has gained urgency across multiple domains. In reinforcement learning, researchers need to determine when agents pursue aligned objectives despite different behaviors and capabilities.[3] AI governance researchers seek to formalize how labor-replacing AI disrupts social contracts and how economic power converts to political influence.[4] While my framework may not immediately solve these challenges, it establishes necessary theoretical foundations.

Methodological Approach. Rather than proposing yet another universal "Theory of Power," we develop rigorous mathematical foundations for one influential perspective – Dahl's concept of power from 1957 – then systematically extend it. This approach allows us to either: (a) formally relate competing definitions to our framework, (b) demonstrate their mathematical incoherence, (c) identify behaviorally irrelevant distinctions, or (d) establish approximation boundaries for complex realities.

As Dahl himself anticipated, we may never achieve "a single, consistent, coherent 'Theory of Power'" but instead must develop "theories of limited scope" that prove useful within specific research contexts.[5]

The Value of Mathematical Formalization. Formal models provide conceptual precision beyond their mathematical machinery. When researchers invoke "prisoner's dilemmas" or "principal-agent problems," they rarely solve explicit equations – instead, they leverage rigorous conceptual frameworks that clarify otherwise murky situations. Similarly, our formal power framework aims to provide precise analytical tools for recognizing and categorizing power dynamics across diverse contexts. The mathematics enables conceptual clarity, not computational solutions.

Our Strategy. We begin with Dahl's 1957 formalization – "A has power over B to the extent that A can get B to do something B would not otherwise do" – because it offers both intuitive appeal and mathematical tractability. We then systematically identify its limitations, demonstrate required extensions, and establish what mathematical properties any complete power theory must satisfy. This constructive approach reveals not just Dahl's shortcomings, but fundamental constraints on power modeling itself.

[1] This reflects a common trade-off in formal modeling: rigor versus generality. Cooperative game theory's power indices (Shapley, Banzhaf, etc.) provide precise mathematical definitions but only apply within specific institutional contexts like weighted voting games 📋 confirm. Conversely, sociological theories achieve broad applicability by sacrificing mathematical precision. The challenge is developing frameworks that maintain both formal rigor and sufficient generality to analyze diverse power structures across different domains.

[2] These sociological approaches capture different but related aspects of power, yet lack mathematical precision needed for quantitative analysis: Dahl's behavioral definition focuses on observable influence over actions but cannot handle indirect influence chains or enable cross-domain power comparisons. French and Raven's taxonomy (reward, coercive, legitimate, referent, expert power) describes power sources but provides no framework for measuring or comparing power magnitudes – it remains unclear whether these categories are irreducible or whether some (e.g., legitimate power) can be decomposed into others (reward/coercive mechanisms). Lukes' three dimensions examine different levels of decision-making processes (overt conflict, covert agenda-setting, ideational preference-shaping) but offer descriptive taxonomies rather than predictive models. Actor-Network Theory treats power as emergent from network relationships but explicitly resists formal quantification, limiting its predictive capacity. While each tradition illuminates important facets—behavioral outcomes, psychological bases, structural processes, relational emergence—none provides the mathematical foundations necessary for rigorous analysis of complex, multi-level power dynamics.

[3] See Jason X, PhD student at Cambridge University

[4] See Liam Patell at GovAI

[5] Dahl's Concept of Power 1957: "Thus we are not likely to produce certainly not for some considerable time to come anything like a single, consistent, coherent 'Theory of Power.' We are much more likely to produce a variety of theories of limited scope, each of which employs some definition of power that is useful in the context of the particular piece of research or theory but different in important respects from the definitions of other studies."

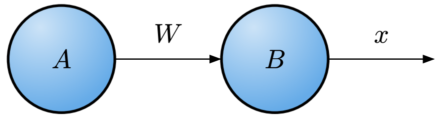


Figure 1. Basic Dahlian Power relationship: Agent A uses means W to influence Agent B's response x.

3. Generalized Dahl Framework

3.1. Defining Dahlian Power

I start with Dahl's influential concept, which I term **Dahlian Power** to distinguish it from other power definitions we'll encounter (Figure 1).

Remark 3.1. (Dahl's Intuition) Consider two scenarios from the original essay[6]

1. A person on a street corner "commanding" drivers to use the right side (which they already do)
2. A police officer directing traffic to turn when it would normally go straight.

Only the second demonstrates genuine power – the ability to cause behavior that wouldn't otherwise occur.

Definition 3.2. *(Dahlian Power)* Agent A's Dahlian Power over B with respect to the response x by means w is:

$$M\left(\frac{A}{B} : W, x\right) := \Pr(B, x \mid A, w) - \Pr(B, x \mid A, \overline{w}) \tag{1}$$

Where:

- $\Pr(B, x \mid A, w)$ is the probability that B takes action x given A took action w.
- $\overline{w}$ is A not taking action w

Remark 3.3. (Key Components) Dahl's model is made up of the following components:

1. **Agents.** At least two agents each of which have binary **actions** (e.g. take action w or not)
2. **Connection.** Some mechanism $\mathcal{J}$ linking agents (unspecified)
3. **Probability.** Distribution over agent-action pairs

What this doesn't require:

1. **Utility.** Doesn't say what either agent prefers.
2. **Causal Relationship.** Dahl explicitly avoids requiring causal traces – only observable behavioral differences matter. However, does note causality is required for real power, hence the means must precede the response temporally.
3. **Practicality.** Doesn't say whether the controller actually would or would not take action w in practice.

3.2. Generalizing Beyond Binary Actions

Dahl's binary framework (action taken or not) is limiting. Real agents choose from multiple options.

Definition 3.1. *(Extension to n Actions)* Let action space $:= \{w_1, ..., w_n \mid n \in \mathbb{N}^+\}$. Then:

$$\begin{aligned} M\left(\frac{A}{B} : W, x\right) := & \max_{w^+ \in \mathcal{W}} \Pr(B, x \mid A, w^+) \\ & - \max_{w^- \in \mathcal{W}} \Pr(B, x \mid A, w^-) \end{aligned} \tag{2}$$

This measures A's maximum influence over B's likelihood of taking action x.

Lemma 3.2. *When* $|\mathcal{W}| = 1$ *(no choice), Dahlian Power* $M = 0$*, fitting our intuition that power requires agency.*

💬 I think Dahl actually notes that you need similar means as well. Look back into this.

3.3. Beyond Two-Controller Dynamics

Real power dynamics rarely involve just two controllers. Consider the example from Dahl's original paper where multiple legislators influence the outcome of a policy being passed or not (Figure 2):

When multiple controllers $\{A_1, ..., A_n\}$ simultaneously influence the responder B, we need to extend our framework. Each agent A_i can take actions from their action space $\mathcal{W}_i$, creating a joint action profile $\boldsymbol{w} = (w_1, ..., w_n)$.

Comparative Dahlian Power. To assess individual contributions within multi-controller dynamics influence, Dahl introduced comparative measures (Referred to as M'' and did not formalize). For controller A_i within controllers A:

📋 Finish, not super important.

[6]The full quote:

"Suppose I stand on a street corner and say to myself, "I command all automobile drivers on this street to drive on the right side of the road"; suppose further that all the drivers actually do as I "command" them to do; still, most people will regard me as mentally ill if I insist that I have enough power over automobile drivers to compel them to use the right side of the road. On the other hand, suppose a policeman is standing in the middle of an intersection a t which most traffic ordinarily moves ahead; he orders all traffic to turn right or left; the traffic moves as he orders it to do. Then it accords with what I conceive to be the bedrock idea of power to say that the policeman acting in this particular role evidently has the power to make automobile drivers turn right or left rather than go ahead. My intuitive idea of power, then, is something like this: A has power over B to the extent that he can get B to do something that B would not otherwise do. – Dahl 1957"

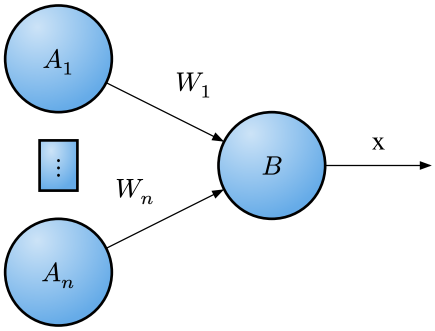


Figure 2. Multi-controller dynamics: Multiple agents $A_1, \ldots, A_n$ simultaneously influence responder B.

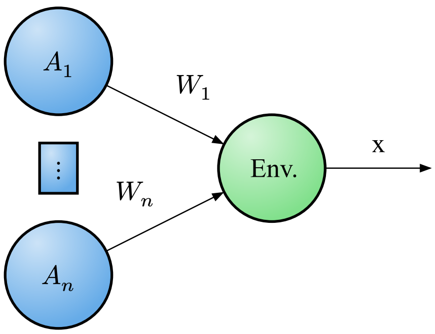


Figure 3. Environment as black box: Multiple agents influence environmental outcomes directly.

$$M_i\left(\frac{A}{B} : \mathcal{W}, x\right) := \max_{w \in \prod \mathcal{W}_j} \ldots \tag{3}$$

Combined Power. We can aggregate any combination of these controllers into a virtual single controller coalition $A' \subseteq \{A_1, \ldots, A_n\}$ and define the power of the coalition in a similar way. This new coalition A' will have an action space constructed from actions spaces of its members $\mathcal{W}' = \mathcal{W}_1 \times \mathcal{W}_2 \times \ldots \times \mathcal{W}_m$ where $A_1, \ldots, A_m$ are the controllers in the coalition. 📋 Better written def, (not necessarily referring to the indices of the original ordering)

3.4. The Environment as a Black Box

Definition 3.1. *(Environmental Power)* Dahl's framework doesn't require the influenced party to be an agent. By treating B's decision-making as a black box abstraction – encoded entirely in the probability function Pr – we can extend Dahlian Power to environmental outcomes (Figure 3). Letting $E :=$ environment:

$$M\left(\frac{A}{E} : W, x\right) := \max_{w^+ \in \mathcal{W}} \Pr(E = x \mid A, w^+) - \max_{w^- \in \mathcal{W}} \Pr(E = x \mid A, w^-) \tag{4}$$

Reverse Environmental Influence. Can environments exert "power"? While environments don't take deliberate actions, environmental states do influence agent behavior (Figure 4):

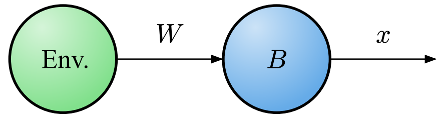


Figure 4. Reverse environmental influence: Environment affects agent behavior through probabilistic states.

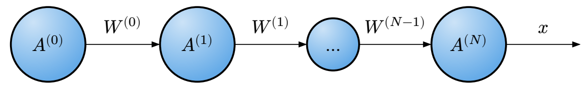


Figure 5. Means-response chain: Sequential influence propagation through intermediate agents.

Example. Weather "determines" whether others will attend your picnic – rain reduces attendance probability. While this stretches our definition of power depending on how you view the environment, it may still be sensible to develop analogues:

- Possible Environmental States = action space

Remark 3.2. (The Agency Question) Dahlian Power doesn't require desires or utility functions – only the ability to select from multiple actions. This creates a philosophical tension:

- Deterministic view: If environments (or agents) have no genuine choice, then $|\mathcal{W}| = 1$, yielding zero Dahlian Power
- Probabilistic view: We assign probabilities to environmental states due to our uncertainty, treating nature as if it "chooses" stochastically

Remark 3.3. (Practical Resolution) Whether discussing weather patterns or legislative decisions, we model both agents and environments probabilistically because:

- Complete determinism is computationally intractable
- Uncertainty is inherent to our observations
- Probabilistic models yield useful predictions

Remark 3.4. (Methodological Choice) **Whether you model environment/people as probabilistic behavior or an agent is a methodological choice, not an ontological claim about free will or determinism.** We use whichever representation – agent or environment – best serves our analytical goals.

3.5. Means-Response Chains and the Limits of Dahlian Power

Consider a means-response chain where A influences B, who influences C, and so on (Figure 5):

Proposition 3.1. *(The Attribution Problem) Dahlian Power fundamentally cannot handle indirect influence chains. To see why, consider:*

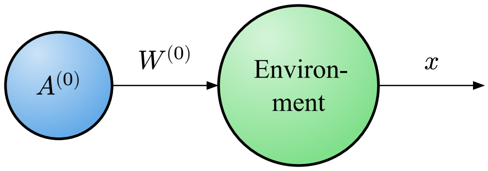


Figure 6. Black-box abstraction: Intermediate chain collapsed into environmental probability function.

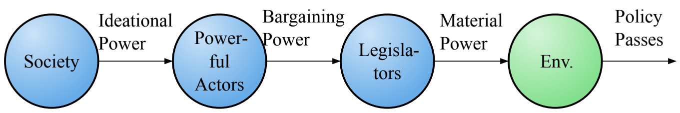


Figure 7. Lukes' critique: Upstream power influences that Dahlian analysis misses in legislative contexts.

- *$A^{(0)}$ has Dahlian Power over $A^{(1)}$'s action $W^{(1)}$*
- *$A^{(1)}$ has Dahlian Power over $A^{(2)}$'s action $W^{(2)}$*
- *But what is $A^{(0)}$'s power over $A^{(2)}$'s action? This could theoretically be determined, yet bears no real relationship to the metrics above. For example: If $A^{(2)}$ has complete ($M = 1$) power over x, then what does any non-zero measure of power over x from $A^{(0)}$ or $A^{(1)}$ mean?*[7]

Proposition 3.2. *(The Black-Box Abstraction Necessity) To analyze $A^{(0)}$'s influence on the final outcome x, Dahl's framework forces us to collapse the entire intermediate chain into an environmental black box abstraction (Figure 6).*

Remark 3.3. (Information Loss) This abstraction loses all structural information about:

- How influence propagates through the network
- Which intermediate agents have veto power
- Where bottlenecks or amplifications occur
- How power dissipates or concentrates along the chain

Theorem 3.4. *(Lukes' Critique Formalized) While Lukes' full critique goes beyond just this issue, this limitation forms one of Lukes' criticisms of Dahlian Power. In studying legislative power, Dahl focused on observable voting behavior, missing upstream influences (Figure 7).*

Remark 3.5. (Power Types Missing from Dahl) Lukes identified two types of power Dahl's framework cannot capture:

1. **Bargaining Power**: Shaping which policies reach the legislative floor
2. **Ideational Power**: Determining which ideas are even consciously considered

[7] This issue has some semblance to the issue of committing a crime under duress in law.

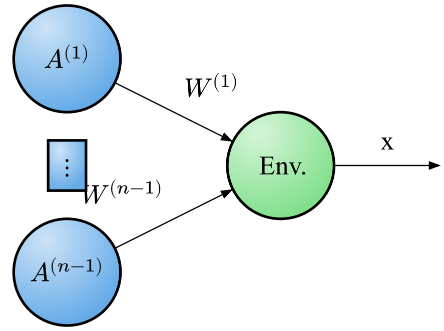


Figure 8. Flattened chain representation: Sequential chain modeled as simultaneous multiple controllers.

These operate through influence chains that Dahlian Power must either ignore or compress into an opaque probability function.

Proposition 3.6. *(Multiple Controllers as Partial Solution) We could remodel the chain as multiple simultaneous controllers (Figure 8).*

Remark 3.7. (Limitations of Flattening) This flattening assumes all agents directly influence the outcome, ignoring:

- Hierarchical relationships (who influences whom)
- Sequential dependencies (temporal ordering matters)
- Conditional influences (power activated only under certain conditions)

The incapability to model these complex relationships is a serious limitation of the model. Lukes' complaint of Dahl's empirical approach is, in part, a complaint that Dahl's power **lacks so much predictive power** by this inability as to be nearly useless. Any serious model of power must move beyond the limitations of this model.

Theorem 3.8. *(The Technical vs. Structural Distinction) This limitation isn't fundamentally about computational complexity. Computer scientists have developed sophisticated inference algorithms for modeling intricate probabilistic relationships – Bayesian networks, Markov networks, loopy belief propagation, and variational inference can handle arbitrary dependency structures within the environmental "black box abstraction." See Figure 9 for an illustration of how a bayes net, starting from each agent's influences on latent variables (which can be thought of as "chains-of-effect"), could model how a final response changes due to some action by a controller.*

Theorem 3.9. *(The Dual Role Problem) The deeper issue is* ***structural****: Dahlian Power's foundational assumption that root agents make decisions independently creates an irreconcilable tension with influence chains. In means-response chains, intermediate agents are simultaneously:*

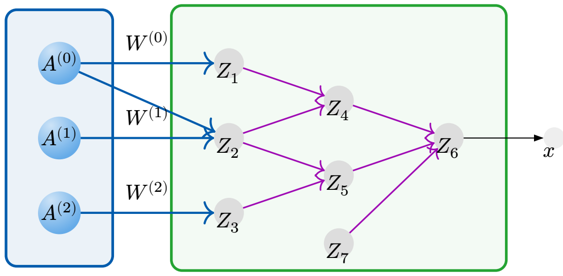


Figure 9. Complex probabilistic structure: Bayesian network showing latent variables and dependencies hidden within Dahl's environmental abstraction.

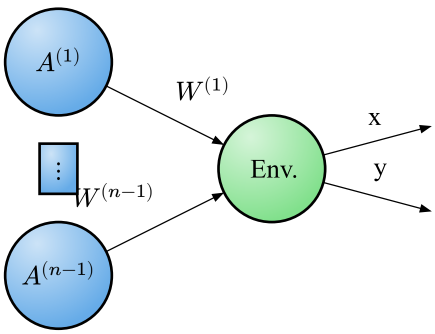


Figure 10. Scope selection limitation: Dahlian Power cannot meaningfully compare power across different response variables.

- ***Responders** to upstream influence (requiring dependent decision-making)*
- ***Controllers** in their own right (requiring independent decision-making)*

This dual role cannot be coherently modeled within Dahl's framework without either:

1. *Abandoning the independence assumption (breaking the foundational logic)*
2. *Collapsing the chain structure (losing predictive information)*

💬 Also modeling causality seems pretty difficult.

3.6. Scope Selection and Limitations

Theorem 3.1. *(Scope Limitation) As discussed in the original essay, Dahlian Power is limited by scope limitation – ie: the response that you're trying to measure (Figure 10). There is not a very sensible way of comparing the power between some response x and some response y.*

Example. From Dahl's original essay:

> "With an average probability approaching one, I can induce each of 10 students to come to class for an examination on a Friday afternoon when they would otherwise prefer to make off for New York or Northampton. With its existing resources and techniques, the New Haven Police Department can prevent about half the students who park along the streets near my office from staying beyond the legal time limit. Which of us has the more power?"
>
> — Dahl's Concept of Power 1957

Remark 3.2. (The Importance Dimension) For a general theory of power, it is extremely limiting to lack the ability to compare power in two separate contexts. While the quote from above was meant to point out how comparing power is difficult and perhaps arbitrary, there are situations for which there is a clear distinction between who has more power in an intuitive sense.

Imagine legislators have different Dahlian Power when voting on two distinct topics: (A) Whether a social welfare program will be approved and (B) Whether town hall's bike shed will be painted green or blue.

Intuitively, if legislator Alice has considerably more Dahlian Power over (A) than legislator Bob, no matter what Bob's Dahlian Power over (B) is, we would still say that Alice has more power. *Our intuitive sense of power relies not only on the ability to influence outcomes, but also how much we care about those outcomes.* This will be revisited later, but for now will be left as a limitation of Dahlian Power.

3.7. General Modeling Difficulties

Remark 3.1. (Practical Modeling Challenges) Dahlian Power has some modeling difficulties that exist across many domains but are nonetheless worth mentioning:

Remark 3.2. (Action Space Selection Problem) Dahlian Power doesn't explain why some actions are included or excluded from the action space, offsetting difficulties to the modeler. E.g. If a legislator can burn down the legislative building instead of voting, guaranteeing the policy doesn't get passed, does the legislator have lots of power? In one sense (the Dahlian Power sense), they do, but in another sense they don't have more power than they did before because the probability they make this decision is essentially 0 – it's up to the modeler to leave this out.

Additionally, it's important to determine which actions are actually available. Partly from Lukes' response, covert bargaining happens before reaching the voting floor. These actions over time and ability to bargain need to be added to the action/strategy space (ex: By defining tuples of actions indicating actions over time, $(\text{bargain for}, \text{vote for})$).

Remark 3.3. (Controller Selection Problem) Partially from Lukes' response, it's important to consider every source of influence that influences the final variable. Ie: It's not just the legislators that determine whether a policy passes,

they're also influenced by society, their family, and others
to different degrees.

4. Why absolute power Fails to Model power Dynamics

📋 Should this only be possible once utility is introduced?

📋 From cooperative game theory, we have weighted graphs, rule-based representations, and weighted-voting games.

4.1. The Modeling Asymmetry Thesis

Proposition 4.1. *Models of relative power ($\mathcal{R}$) form a proper superset of models of absolute power ($\mathcal{A}$), establishing that $\mathcal{R}$ is more expressive than $\mathcal{A}$ in modeling capacity.*

Dahl says that there must be a "connection" for there to be a power dynamic.

4.1.1. Formal Statement

Definition 4.2. *(Power Model Spaces)* Let:

- $\mathcal{A} = \{M : M \text{ models power as properties of agents}\}$
- $\mathcal{R} = \{M : M \text{ models power as relational properties between agents}\}$

Then: $\mathcal{A} \subset \mathcal{R}$ (proper subset relation)

4.1.2. Supporting Arguments

Theorem 4.3. *(Reduction Principle) Any absolute power model can be reformulated as a relative power model by introducing an environmental baseline.*

Formal Construction. Given absolute power P_{abs} : Agent $\to \mathbb{R}^+$, we can define:

$$P_{\text{rel}(a,e)} = P_{\text{abs}(a)} - P_{\text{abs}(e)} \tag{5}$$

where $P_{\text{abs}(e)}$ serves as a zero-point reference for environment e.

📋 Replace with an actual proof from graph theory

Physical Analogy. This construction mirrors potential energy in physics, where absolute potential V is meaningless without a reference frame:

$$V_{\text{rel}} = V_{\text{abs}} - V_{\text{ground}} \tag{6}$$

Incompleteness of Absolute Power Models ($\mathcal{A} \subsetneq \mathcal{R}$).

Theorem 4.4. *(Limitation Theorem) Absolute power models cannot capture essential relational dynamics without introducing factors outside their framework.*

📋 Demonstrate some examples like in my original, such as distance as one factor and just other seemingly complex set of environmental interaction factors that seem almost impossible to resolve absolute powers between entities to relative powers in a sensible way. In physics potential can be mapped to potential energy using distance, unlikely we have anything nearly as similar.

Core Problem. The power relationship between agents A and B cannot be logically derived from their absolute powers alone. Power lacks transitivity – if A dominates B and B dominates C, this does not guarantee A dominates C. This non-transitivity demonstrates that power cannot be reduced to a single absolute measure.

Example. Consider a circular dominance structure (like rock-paper-scissors):

- Rock defeats Scissors
- Scissors defeats Paper
- Paper defeats Rock

No absolute power assignment P_{abs} can capture these relationships.

Conclusion. Since capturing these dynamics requires factors $\notin \mathcal{A}$, we have $\mathcal{A} \neq \mathcal{R}$.

📋 improve

4.1.3. Proof of Proper Subset Relation

From §1.2.1: $\mathcal{A} \subseteq \mathcal{R}$
From §1.2.2: $\exists M \in \mathcal{R}$ such that $M \notin \mathcal{A}$
Therefore: $\mathcal{A} \subset \mathcal{R}$ (proper subset) □

4.1.4. Power Over Environment as Logical Encapsulation of What is Meant By "Absolute Power"

📋 fin

4.1.5. Conclusion

Absolute power alone is insufficient for modeling power. For the remainder of this work, we consider only relational power – power over the environment or power over other agents.

Note: This does NOT tell us that relative power is sufficient, only that it may be.

5. Why Utility is Necessary for Power Analysis

5.1. The Inseparability Thesis

Proposition 5.1. *Power is meaningless without considering agent utilities – these elements are fundamentally inseparable in any coherent power framework.*

NOTE: Will likely replace this utility statement.

5.1.1. Utility Alignment

Core Insight. Power is an incoherent concept under complete alignment of utilities.

Definition 5.2. *(Utility Alignment)* Let:
- $U_A, U_B : \mathcal{S} \to \mathbb{R}$ be utility functions for agents A and B
- $\mathcal{S}$ be the state space
- $P(A \to B)$ be the power of A over B

If $U_A(s) \propto U_B(s)\ \forall s \in \mathcal{S}$ (perfect alignment), then $P(A \to B)$ is behaviorally irrelevant.

Illustrative Example. Consider a human-loving agent who can travel to an island, harm anywhere from 1 to 1,000,000 innocent people, and then escape unnoticed. Does their power increase with the number they *could* harm? This capacity is irrelevant:
- The human-loving agent doesn't benefit from causing harm and would prefer not to
- The potential victims have no desire to be harmed

The mere existence of this action is meaningless without considering whether it would ever be exercised.

📋 Replace with better example.

📋 Include behavioral misalignment, ie: power over

5.1.2. Agent Boundary Definition

Theorem 5.3. *(Aggregation Principle) Alignment relationships determine when individual power can be meaningfully aggregated into a single coherent agent:*

$$\left[P_{\text{collective}} = \sum_i P_i\right] \Leftrightarrow [U_i \approx U_j\ \forall i, j] \tag{7}$$

Application. Humanity's collective power is discussable precisely because humans share sufficient utility alignment over key domains.

Connection to Sociology This is akin to *power-within* discussed in sociological literature.

5.1.3. Environmental Mediation

Extended Framework. When modeling multi-agent systems with environmental interactions:
- Environment E can be treated as an agent with its own dynamics
- Even without direct conflict between agents, they may compete for environmental resources
- Agents may inflict harm indirectly through environmental manipulation

Power becomes relevant when agents compete for control over shared environmental factors, even if their ultimate goals don't directly conflict.

📋 Also could say that their actions are some mixed strategy over a number of actions. Not sure if this is true but not a crucial point.

5.1.4. Agents in Shared State Space

Agents of course, are themselves components of state spaces and they may have preferences over the state of themselves and others. Often times they have preferences for their survival which must be modeled in the utility function as well.

5.1.5. Critique of Light-Cone Formulation

Orthogonality Scenario. Consider alien civilization Ω with utilities orthogonal to humanity H. Imagine Ω cares only about arranging neutrinos into smiley faces – an activity that has zero impact on human flourishing. Meanwhile, humans optimize matter and energy arrangements with no effect on neutrino patterns. (We can additionally imagine that neither agent meaningfully interferes with the other, perhaps humanity and these aliens never interact)

Even if Ω achieves perfect control over neutrino arrangements while humans have only partial control over matter/energy, it's meaningless to compare their "power over the environment." The concept of environmental control is incoherent without specifying *which aspects* of the environment matter.

Key Insight. Power must be defined over relevant state space – control over factors that actually matter to the agents involved:

$$P_{\text{relevant}(A)} = \int_{\mathcal{S}_A} |\nabla U_A(s)|\ \ ds \tag{8}$$

where $\mathcal{S}_A := \{s \in \mathcal{S} : \partial U_A / \partial s \neq 0\}$ represents the relevant state space.

Note that this relevant state space is inherently chosen in reference to some agent and thus is incoherent outside of some utility reference frame.

5.2. Synthesis

Theorem 5.1. *(Completeness Requirement) A complete theory of power requires (although may be insufficient):*

1. *Relational framework ($\mathcal{R} \supset \mathcal{A}$)*
2. *Behavioral specifications for all agents (possibly without explicit utilities)*
3. *Action space definitions $\mathcal{A}_i$*
4. *Relevance constraints on state space $\mathcal{S}_i \subset \mathcal{S}$*

Implication. Power analysis without game-theoretic foundations is fundamentally incomplete. □

📋 Power of signaling. US vs Japan and communicating the existence of the nuclear bomb. US in some way doesn't have power over Japan if it doesn't benefit from either (A) Annihilating it or (B) Communicating its Annihilation.

References

A. Glossary

absolute power: Power conceptualized as an intrinsic property of individual agents, independent of relationships. We prove this approach is insufficient for modeling power dynamics. 1, 1, 8

action space: The set of possible actions $\mathcal{W} = \{w_1, ..., w_n\}$ available to an agent. Power requires $|\mathcal{W}| > 1$ (genuine choice between alternatives). 3, 3, 4, 4, 6

agent: An entity capable of taking actions from a defined action space. In our framework, agents can serve as either controllers (exerting influence) or responders (being influenced). 1, 1, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5

black box abstraction: Modeling complex intermediate processes (like means-response chains) as environmental probability functions, losing structural information about how influence propagates. 4, 5, 5

controller: An agent that attempts to influence the behavior of other agents (responders) through its actions. The agent A in a Dahlian power relationship $\frac{A}{B}$. 3, 3, 3, 3, 3, 4, 4, 4, 5, 5, 6

Dahlian Power: Power as formalized by Dahl (1957): $M\left(\frac{A}{B} : W, x\right) := \Pr(B, x \mid A, w) - \Pr(B, x \mid A, \overline{w})$. Measures the difference in probability of response x when agent A takes action w versus not taking action w. 1, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6, 6

environment: A non-agential entity that responds probabilistically to agent actions without deliberate decision-making. Represented by probability distributions $\Pr(E = x \mid A, w)$ rather than utility-maximizing behavior. 4, 4, 4, 4, 4, 4, 4

means-response chain: A sequence of agents where each serves as both responder to upstream agents and controller of downstream agents: $A^{(0)} \to A^{(1)} \to ... \to A^{(N)}$. Cannot be modeled coherently within Dahlian framework. 4, 5

multi-controller dynamics: Scenarios where multiple agents $\{A_1, ..., A_n\}$ simultaneously influence a single responder B, requiring analysis of joint action profiles $\boldsymbol{w} = (w_1, ..., w_n)$. 3

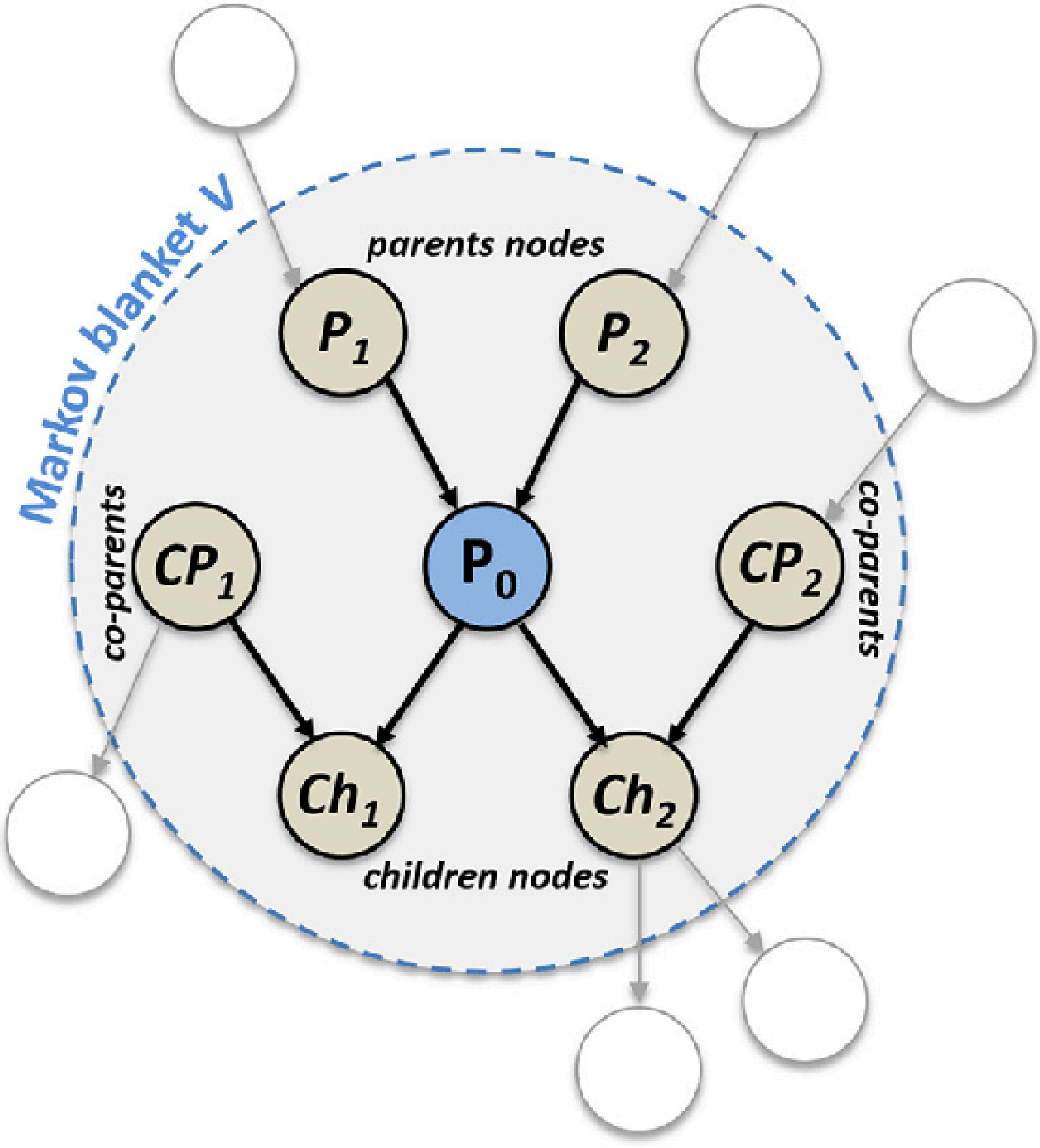
power: The intuitive notion of influence, control, or ability to affect outcomes. While this term encompasses many different conceptions across disciplines, we focus on developing rigorous mathematical foundations for analyzing power relationships between agents. 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 8

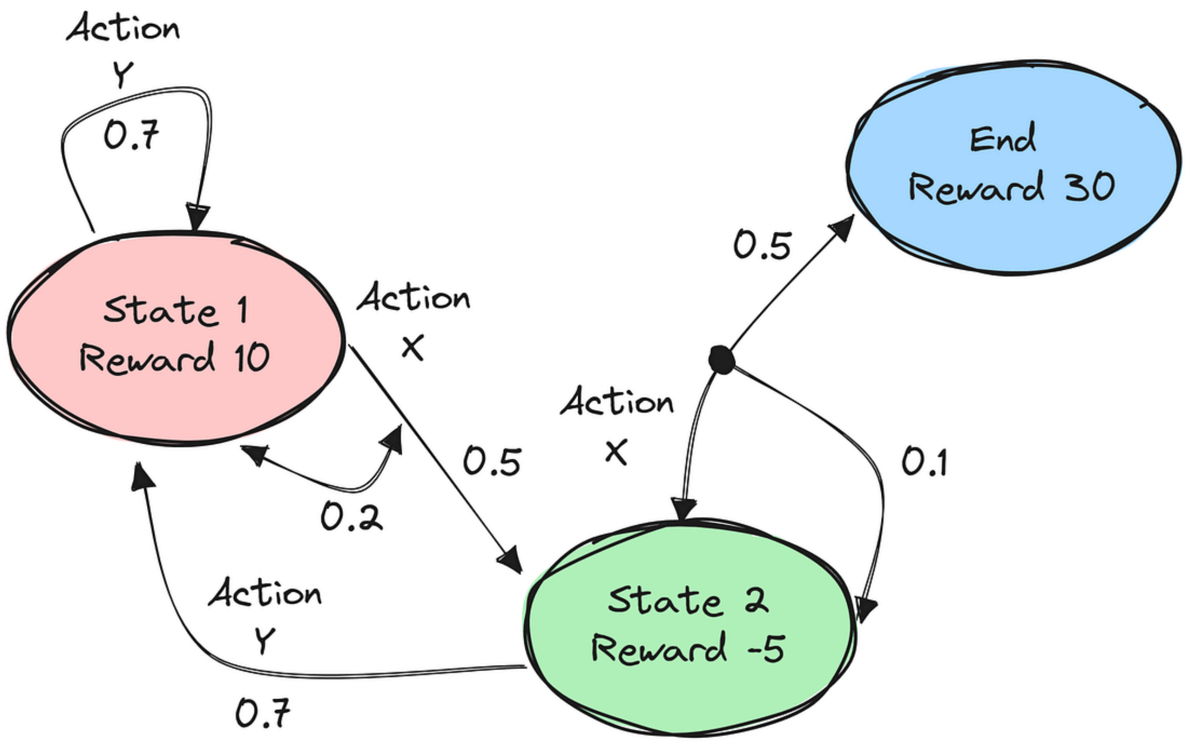
relative power: Power conceptualized as relational properties between agents. Forms a proper superset of absolute power models and is necessary for analyzing complex power dynamics. 1

responder: An agent whose behavior is potentially influenced by controllers. The agent B in a Dahlian power relationship $\frac{A}{B}$. Can be modeled as either an agent or environmental black box. 3, 6

scope limitation: Dahlian power's inability to compare power across different response variables (e.g., power over policy A versus power over policy B), limiting cross-domain power analysis. 6

B. Generalizing Dahl to Markov Nets, HMMs, and MDPs


Markov blanket V
parents nodes
P_1
P_2
co-parents
CP_1
P_0
CP_2
co-parents
Ch_1
Ch_2
children nodes



In broad strokes, Dahl's theory of power is similar to the rich literature of markov nets in computer science literature. With a few differences:

- Taken from the perspective of actions that players take. (However these can be mapped onto priors in a markov net so long as there are no loops, which of course there are. Might not actually be an issue.)
- Control must be causal. Since this is near-impossible to prove in-full, it suffices to say that it's certainly not causal if the response happens before the action.

A few nice properties and intuitions can however can be carried over:

- Markov blankets (You can isolate control by)
- Can be mapped to time-series HMMs.

Notoriously very difficult. 📋 Maybe also show some existing CS problem to apply control in a new context. 📋 Also maybe see what other studies of power exist out there that looked at causal relationships in this way. Certainly not a new understanding of power, just one that wasn't looked at broadly.

Already we are at a point where we can define control in MDPs, one of the most fundamental structures in reinforcement learning and applicable to many scenarios. 📋 Can we apply this to some textbook MDP scenario? 📋 Not sure I should transition to MDPs without bringing in utility as a pre-requisite of power.